Component Diversity, Excess Entropy, and Divalent-Ion Self-Diffusion in Charged Lennard-Jones Mixtures

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Does increasing the number of mixture components raise the excess entropy of a liquid and accelerate the divalent ions dissolved in it? We address this question for charged Lennard-Jones mixtures at $T^* = 2$, $\rho^* = 0.7$ that contain fixed fractions, 0.03 per sign, of $+2/-2$ probe ions. Hypernetted-chain (HNC) integral-equation theory is used to screen 35 candidate compositions, and eight selected mixtures are studied by independently replicated molecular dynamics. The excess entropy is obtained by thermodynamic integration over 19 densities with a fixed shifted LJ cutoff and a Debye–Hückel dilute endpoint; transport is measured in a single common center-of-mass frame. Solvent designs selected for high excess entropy raise it by approximately $0.30k_B$ per particle and increase divalent self-diffusion by 18–20% relative to the baseline electrolytes, with both ion signs accelerating. A controlled diversification, which preserves the charged composition and the mean bare LJ pair amplitude, raises the excess entropy by $0.103k_B$ and the mean self-diffusion by 8–9%, while the probe ions respond more weakly. Six additional matched replicas of 5000τ resolve this probe response as +5.1% without and +5.9% with monovalent background, unchanged in systems twice as large. The mean self-diffusion follows the accompanying drop in shear viscosity, whereas the divalent ions are only partly carried along. Repeating the study at 0.075 per sign, a total divalent fraction of 0.15, reproduces all entropy and diffusion contrasts within their uncertainties. The Green–Kubo ionic conductivity is a composition-independent 73% of its Nernst–Einstein value, so it follows the number of carriers and their self-diffusion. Splitting a solvent into mechanically identical labels changes the ideal mixing entropy but leaves diffusion exactly unchanged, and adding monovalent salt alone does not accelerate the probes. Pair entropy computed from the MD correlations accounts for about 80% of the excess entropy and of its controlled increments. The results identify faster-ion, higher-excess-entropy mixtures and a positive Rosenfeld-like trend for both the mean and the probe diffusion, but establish neither a causal effect of component count nor a universal species scaling law.

I. INTRODUCTION

Excess-entropy scaling offers a link between equilibrium correlations and liquid transport. Rosenfeld proposed that an appropriately reduced diffusion coefficient obeys [1, 2]

$$D_i^R = A_i \exp(B_i s_{\rm ex}/k_B), \qquad D_i^R = D_i \rho^{1/3}\sqrt{m_i/(k_B T)}. \tag{1}$$

For mixtures, whether this relation describes individual charged species as well as the solvent-dominated mean is a separate question. Local solvation, association, and interaction-dependent friction can respond differently to composition. Rosenfeld scaling has been applied to binary dense-plasma mixtures [3], while generalized tracer formulations show that mixture diffusion can depend on kinetic information beyond a single bulk entropy coordinate [4]. Recent work has tested entropy scaling across a broad range of fluids and mixtures [5].

The entropy entering Eq. (1) is the excess per particle relative to the ideal mixture at the same temperature, density, and composition. The ideal reference already includes $s^{\rm id}_{\rm mix}/k_B = -\sum_i x_i \ln x_i$. A greater number of species can therefore increase the total entropy without increasing $s_{\rm ex}$ or changing any force. Distinguishing label multiplicity from physical interaction diversity is essential to any test of a component-induced speedup.

The question has practical relevance. In battery electrolytes, high-entropy formulations have been associated with enhanced lithium transport [6], and a Li-centered pair-entropy proxy has been correlated with Li diffusivity across carbonate-electrolyte formulations [7]. Such local pair statistics differ from the total thermodynamic excess entropy. These observations motivate a controlled model study in which ideal mixing, total excess entropy, and local pair structure are separated.

Here we use HNC theory to select charged Lennard-Jones mixtures with high excess entropy, and test the selection by independently replicated molecular dynamics (MD). The design includes an exact label-only null comparison and physically diverse solvents at matched mean bare interaction amplitude. We distinguish the particle-averaged self-diffusion $D_{\rm net} = \sum_i x_i D_i$ from mutual diffusion and conductivity. The divalent quantity is the average self-diffusion of the two probe signs, not the motion of neutral ion pairs. The central tests are whether physical diversification raises the MD excess entropy, whether the probes accelerate reproducibly, and whether their response is as strong as that of the mean fluid.

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II. MODEL AND THEORY

A. Interaction Potentials

We consider a classical mixture of p components with number densities $\rho_i = N_i/V$, total density $\rho = \sum_{i=1}^{p} \rho_i$, and mole fractions $x_i = \rho_i/\rho$. The particles interact via a cut-and-shifted Lennard-Jones (LJ) core augmented by Coulomb interactions:

$$u_{ij}(r) = u_{ij}^{\mathrm{LJ}}(r) + \frac{q_i q_j e^2}{4\pi\varepsilon_0\varepsilon_r r}, \tag{2}$$

where the LJ contribution is truncated and shifted at $r_c = 2.5\,\sigma$:

$$u_{ij}^{\mathrm{LJ}}(r) = \begin{cases} 4\varepsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r}\right)^{12} - \left(\frac{\sigma_{ij}}{r}\right)^{6}\right] - u_{ij}^{\mathrm{shift}}, & r \le r_c, \\ 0, & r > r_c. \end{cases} \tag{3}$$

All particles share identical diameters $\sigma_{ij} = \sigma$ and masses $m_i = m$. Unlike dispersion energies follow the geometric rule $\varepsilon_{ij} = \sqrt{\varepsilon_i \varepsilon_j}$. Reduced LJ units are used throughout: $r^* = r/\sigma$, $\rho^* = \rho\sigma^3$, $T^* = k_B T/\varepsilon_1$, $u^* = u/\varepsilon_1$, and $t^* = t/\tau$ with $\tau = \sigma\sqrt{m/\varepsilon_1}$.

The reduced Coulomb energy prefactor is $\Gamma = e^2/(4\pi\varepsilon_0\varepsilon_r\sigma\varepsilon_1) = 1$, so the reduced Bjerrum length is $\lambda_B^* = \Gamma/T^* = 0.5$ at $T^* = 2$. The ionic strength and Debye screening parameter are

$$I = \frac{1}{2}\sum_{i=1}^{p} x_i q_i^2, \qquad \kappa = \sqrt{8\pi\lambda_B^* \rho^* I}. \tag{4}$$

B. Ornstein–Zernike Equations and the HNC Closure

The structure of the mixture is described by the total correlation functions $h_{ij}(r) = g_{ij}(r) - 1$ and the direct correlation functions $c_{ij}(r)$, related by the multicomponent Ornstein–Zernike (OZ) equation [8]

$$h_{ij}(r) = c_{ij}(r) + \sum_{k=1}^{p} \rho_k \int c_{ik}(|\mathbf{r} - \mathbf{r}'|) h_{kj}(r')\, d\mathbf{r}', \tag{5}$$

which in Fourier space takes the matrix form

$$\tilde{\mathbf{H}}(k) = \tilde{\mathbf{C}}(k) + \tilde{\mathbf{C}}(k)\mathbf{P}\tilde{\mathbf{H}}(k), \qquad \mathbf{P}_{ij} = \rho_i \delta_{ij}. \tag{6}$$

The hypernetted-chain (HNC) closure, introduced by Morita and Hiroike through a diagrammatic resummation of the pair distribution function [9] and extended to multicomponent systems by Hiroike [10], reads

$$g_{ij}(r) = \exp\left[-\beta u_{ij}(r) + h_{ij}(r) - c_{ij}(r)\right], \tag{7}$$

and corresponds to neglecting the bridge function. For charged particles, both $c_{ij}(r)$ and $h_{ij}(r)$ inherit the long range of the Coulomb interaction. Following Allnatt [12], the OZ equation is renormalized so that the Coulomb chain sum is performed analytically and only short-ranged functions remain. We use the explicit implementation of Lomba and Høye [13]. The long-range part of the direct correlation function is represented by a Coulomb tail regularized at the origin by a Yukawa factor,

$$c_{ij}(r) = c_{ij}^{\mathrm{sr}}(r) + \psi_{ij}(r), \qquad \psi_{ij}(r) = -\frac{\lambda_B^* q_i q_j}{r}\left(1 - e^{-\xi r}\right), \tag{8}$$

where the inverse range ξ is tied to the Debye parameter κ. The corresponding chain sum is known in closed form. It yields a difference of two screened Yukawa functions $q_{ij}(r)$, with decay constants fixed by ξ and κ, such that $h_{ij}^{\mathrm{sr}} = h_{ij} - q_{ij}$ is also short ranged. The OZ equation is then a matrix equation in k space that involves only short-ranged functions at any density, and the closure, Eq. (7), is evaluated with c^{sr}, h^{sr}, and the analytic tails. The equations are solved by a hybrid Newton–Raphson scheme [13, 14]. Newton–Raphson iterations with an analytic Jacobian are applied to the low-k components of the indirect correlation function $\tilde{h}^{\mathrm{sr}} - \tilde{c}^{\mathrm{sr}}$, which carry most of the structure. The remaining components are updated by direct iteration, and successive iterates are mixed adaptively. Following Kinoshita and Harada [15], the Jacobian is reused along a sequence of state points, which makes density and temperature sweeps efficient. The original binary charged-hard-sphere program [13] is generalized here to an arbitrary number of charged LJ components and applied on a radial grid of $N_r = 4096$ points with $\Delta r = 0.005\,\sigma$.

C. Thermodynamics and Excess Entropy

The excess internal energy and compressibility factor follow from the energy and virial routes,

$$\frac{u_{\mathrm{ex}}}{k_B T} = \frac{\beta\rho}{2}\sum_{i,j} x_i x_j \int 4\pi r^2 u_{ij}(r) g_{ij}(r)\, dr, \tag{9}$$

$$Z = 1 - \frac{\beta\rho}{6}\sum_{i,j} x_i x_j \int 4\pi r^3 \frac{du_{ij}(r)}{dr} g_{ij}(r)\, dr. \tag{10}$$

A distinctive property of the HNC approximation is that the excess Helmholtz free energy is a closed functional of the pair correlations, without coupling-constant or density integration. This result follows from the variational formulation of Morita and Hiroike [11]; for a mixture, with $a_{\mathrm{ex}} = A_{\mathrm{ex}}/(N k_B T)$ [11, 16],

$$\begin{aligned} a_{\mathrm{ex}}^{\mathrm{HNC}} = &-\frac{1}{2\rho}\sum_{i,j}\rho_i\rho_j \int d\mathbf{r}\left[\frac{1}{2}h_{ij}^2 - c_{ij}h_{ij}\right. \\ &\left. + h_{ij} - (h_{ij}+1)\ln(h_{ij}+1)\right] \\ &- \frac{1}{2\rho(2\pi)^3}\int d\mathbf{k}\Big[\ln\det(\mathbf{I} + \mathbf{P}^{1/2}\tilde{\mathbf{H}}\mathbf{P}^{1/2}) \\ &- \mathrm{Tr}\left(\mathbf{P}\tilde{\mathbf{H}}\right)\Big]. \end{aligned} \tag{11}$$

Within HNC, the pressure obtained from this free energy coincides with the virial pressure, Eq. (10). Their agreement therefore checks the internal consistency of the numerical solution [13]. The excess entropy per particle then follows as

$$\frac{s_{\rm ex}}{k_B} = \frac{u_{\rm ex}}{k_B T} - a_{\rm ex}. \tag{12}$$

In MD the same quantity is obtained by integrating the equation of state along the isotherm, $a_{\rm ex}(\rho) = \int_0^\rho [Z(\rho') - 1]\, d\rho'/\rho'$ (Sec. III B).

D. Pair Structural Entropy

The two-body contribution to the excess entropy is [17, 18]

$$\frac{s_2}{k_B} = -\frac{\rho}{2} \sum_{i,j} x_i x_j \int \left[g_{ij}(r) \ln g_{ij}(r) - g_{ij}(r) + 1 \right] d\mathbf{r}. \tag{13}$$

The integrand vanishes as $g_{ij} \to 1$, so the integral converges rapidly. s_2 measures the loss of translational entropy due to static pair correlations [18]; it is evaluated both from HNC and directly from the MD pair distributions.

III. COMPOSITION DESIGN AND SIMULATION METHODS

A. Composition Design and Controls

All mixtures contain $x_{+2} = x_{-2} = 0.03$, with equal masses and nominal diameters, at $T^* = 2$ and $\rho^* = 0.7$. Table I defines the eight states. B3 is a probe electrolyte in a single neutral LJ solvent, and B5 adds 5% of each monovalent sign to it. A second campaign at $x_{+2} = x_{-2} = 0.075$ is described in Sec. III E.

Thirty-five integer-charge, mole-fraction, and dispersion candidates were screened with HNC on 2048 radial points with $\Delta r = 0.01\sigma$; all converged. The eight selected candidates were refined on 4096 points with $\Delta r = 0.005\sigma$, and their excess entropies agree with the coarse results to $10^{-4} k_B$. The low-wavevector partial structure-factor matrices were positive for these states. This is a screening diagnostic, not a proof of phase stability. Mixtures of conjugate charged species did not provide a large positive excess-entropy signal, so the replicated MD effort concentrates on solvent diversification and on the salt-addition comparison.

U4 contains one secondary neutral solvent with $\varepsilon = 0.25$. L5 splits that solvent into equal populations of mechanically identical labels. M5 and H5 instead split it into two physically distinct solvents with ε pairs $(0.0625, 0.5625)$ and $(0.01, 0.81)$. Under geometric mixing, all four states have

$$\sum_i x_i \sqrt{\varepsilon_i} = 0.85, \qquad \sum_{ij} x_i x_j \varepsilon_{ij} = 0.85^2 = 0.7225. \tag{14}$$

Their ion fraction, ionic strength, nominal sizes, and masses are also identical. U6 and H7 repeat the uniform/diverse comparison with 5% of each monovalent sign, at the same mean bare pair amplitude. Equation (14) does not fix the actual potential energy, local solvation, effective excluded volume, or friction: the distribution of interactions still changes. H5 and H7 are deliberately strong perturbations, while M5 is a moderate counterpart.

B. Simulation and Entropy Integration

LAMMPS simulations [19] use $N = 4000$, including 120 probes of each divalent sign. Each composition has three independent replicas. Within each replica index, compositions share initial positions and probe identities, enabling paired comparisons. Species are assigned at random on a perturbed cubic lattice, followed by energy minimization. All particles are integrated with a single Nosé–Hoover NVT thermostat, damping time 0.1τ, and timestep 0.002τ.

Each replica covers 19 densities:

$$\begin{aligned} \rho^* = {} & 0.001, 0.0025, 0.005, 0.01, 0.02, 0.05, \\ & 0.10, 0.15, 0.20, 0.25, 0.30, 0.35, \\ & 0.40, 0.45, 0.50, 0.55, 0.60, 0.65, 0.70. \end{aligned} \tag{15}$$

Every state uses the same LJ potential, cut and shifted at 2.5σ, without tail correction. Electrostatics use PPPM with relative accuracy 10^{-5} and a real-space Coulomb cutoff $5(0.7/\rho^*)^{1/3}\sigma$, below half the box length. Only the numerical electrostatic splitting changes with density; the Hamiltonian is the same along the whole integration path. The 18 lower-density states use 50 000 equilibration and 100 000 production steps. At $\rho^* = 0.7$, 100 000 equilibration steps precede 500 000 production steps (1000τ); this transport run also supplies the final thermodynamic point.

Entropy integration uses the final half of production at each density. The all-particle temperature has $3N - 3$ kinetic degrees of freedom after removal of the total momentum. The configurational compressibility factor is therefore

$$Z_{\rm ref} = 1 + \frac{\langle P \rangle - \rho(1 - 1/N)\langle T \rangle}{\rho T_0}, \qquad T_0 = 2, \tag{16}$$

which removes the sampled kinetic contribution and its finite-N offset from the excess equation of state. It does not remove finite-size effects in configurational correlations. With $y = \sqrt{\rho^*}$,

$$a_{\rm ex}(\rho^*) = \int_0^{\sqrt{\rho^*}} \frac{2[Z_{\rm ref}(y^2) - 1]}{y}\, dy, \tag{17}$$

TABLE I. Selected replicated MD systems. Every mixture contains $x_{+2} = x_{-2} = 0.03$ and has unit masses and nominal diameters. The primary neutral solvent and all ions have $\varepsilon = 1$. Fractions of secondary neutral solvents and their respective ε values are listed in matching order; $x_{\pm1}$ is the fraction of each monovalent sign. Label count and mechanically distinct species count are separated. $I = \frac{1}{2}\sum_i x_i q_i^2$.

State	Labels/distinct	$x_{\varepsilon=1,\,q=0}$	Secondary fractions	Secondary ε	$x_{\pm1}$	I
B3	3/3	0.94	—	—	0	0.12
B5	5/5	0.84	—	—	0.05	0.17
U4	4/4	0.64	0.30	0.25	0	0.12
L5	5/4	0.64	0.15+0.15	0.25, 0.25	0	0.12
M5	5/5	0.64	0.15+0.15	0.0625, 0.5625	0	0.12
H5	5/5	0.64	0.15+0.15	0.01, 0.81	0	0.12
U6	6/6	0.54	0.30	0.25	0.05	0.17
H7	7/7	0.54	0.15+0.15	0.01, 0.81	0.05	0.17

whose integrand has the finite Debye–Hückel limit

$$F(0) = -\frac{[4\pi\lambda_B^* \sum_i x_i q_i^2]^{3/2}}{12\pi}. \tag{18}$$

A cubic spline through this endpoint and the simulated densities gives $a_{\rm ex}$, and $s_{\rm ex}/k_B = \langle U_{\rm ex}\rangle/(Nk_BT_0) - a_{\rm ex}$. Shape-preserving interpolation and alternative averaging windows test the sensitivity. The ideal reference is always the ideal mixture at the same composition, so its mixing entropy is excluded from $s_{\rm ex}$.

C. Transport Estimators and Uncertainty

Unwrapped positions are recorded every 0.5τ for the full dense production run. The same whole-system center-of-mass displacement is subtracted from every particle. Multiple time origins provide species mean-squared displacements (MSD) up to 250τ. A linear fit over 20–100τ, divided by six, is primary; 10–50τ and 50–200τ are sensitivity windows. The identity $D_{\rm net} = \sum_i x_i D_i$ is checked at every lag. Since both probe populations are equal, $D_{\rm probe} = (D_{+2} + D_{-2})/2$. Velocities sampled every 0.01τ during the final 100τ provide species velocity autocorrelation functions (VACF) as a check, and low-wavevector concentration fluctuations are monitored as a homogeneity diagnostic.

Uncertainty comes from independent replicas, not overlapping MSD origins. Composition effects use paired entropy differences and paired $\ln(D_a/D_b)$, with nominal 95% Student-t intervals. Percent changes and interval endpoints follow by exponentiation. No multiple-comparison correction is applied, and finite-size and estimator biases are outside these intervals. L5 is an exact mechanical duplicate of U4 and is never counted as an independent state in scaling fits.

D. Extended Sampling and Box-Size Campaigns

Two further dense-state campaigns test the controlled ion response. The first adds six new independent replicas of B3, U4, H5, U6, and H7 at $N = 4000$, matched across compositions within each replica, with 5000τ of production after 200τ of equilibration. The second repeats the same five states with $N = 8000$ (240 probes of each sign) in four matched replicas of the same length. Interactions, cutoffs, PPPM accuracy, thermostat, and timestep are unchanged. Positions are stored every 1τ. The MSD is fitted over the three windows above and additionally over 100–500τ and 200–1000τ. The off-diagonal and traceless diagonal pressure-tensor components, sampled every 0.01τ, give the shear viscosity

$$\eta = \frac{V}{k_BT}\int_0^\infty \langle P_{\alpha\beta}(0)P_{\alpha\beta}(t)\rangle\, dt, \tag{19}$$

averaged over the five independent components. Its running integral changes by less than 1.5% between 5 and 20τ. The viscosity enters the hydrodynamic finite-size correction of Yeh and Hummer [20],

$$D_\infty = D_L + \frac{2.837297\, k_BT}{6\pi\eta L}, \tag{20}$$

where L is the box length.

E. Higher Divalent Concentration and Ionic Conductivity

To test whether the conclusions hold at ion contents closer to those of electrolytes, the whole design was repeated with $x_{+2} = x_{-2} = 0.075$, a total divalent fraction of 0.15 (300 ions of each sign at $N = 4000$). The primary-solvent fraction of every composition is reduced by 0.09, while all secondary-solvent and monovalent fractions and all LJ parameters are kept. Equation (14) therefore still holds for U4, L5, M5, and H5, and U6 and H7 again form a matched pair. All 35 HNC candidates were rescreened and converged. The HNC solutions also converged at all 19 densities of the integration path, with positive low-wavevector structure-factor eigenvalues. At $\rho^* = 0.7$, HNC solutions for H5 and H7 continue down to $T^* = 1.25$ and for M5 to $T^* \simeq 1.1$, below which the closure fails under the stronger divalent coupling. This is a limit of the

HNC solution, not a phase boundary. All MD remains at $T^* = 2$.

The eight states were simulated in four matched replicas with the 19-density protocol of Sec. III B. The only change is that the dense run has 3000τ of production. Positions are stored every 1τ, and the pressure tensor and the total charge current $\mathbf{J} = \sum_i q_i \mathbf{v}_i$ every 0.01τ. The Green–Kubo ionic conductivity is

$$\sigma = \frac{1}{3Vk_BT} \int_0^\infty \langle \mathbf{J}(0) \cdot \mathbf{J}(t) \rangle \, dt, \tag{21}$$

in reduced LJ units. Both σ and η are taken as the mean of their running integrals between 5 and 10τ. The conductivity is compared with the Nernst–Einstein value $\sigma_{\mathrm{NE}} = (\rho/k_BT) \sum_i x_i q_i^2 D_i$, built from the measured self-diffusion coefficients. Their ratio $\sigma/\sigma_{\mathrm{NE}}$, the inverse Haven ratio, measures the cross correlations between the motions of different ions.

IV. RESULTS AND DISCUSSION

A. Higher Excess Entropy Selected by HNC

Table II reports the dense-state results, and Table III the corresponding absolute-state intervals. HNC successfully selects higher-excess-entropy liquids: H5 relative to U4 has $\Delta s_{\mathrm{ex}} = +0.1107 k_B$ in HNC and $+0.10322 k_B$ in MD. With monovalent background, H7 relative to U6 gives $+0.1108 k_B$ and $+0.10331 k_B$, respectively. Absolute HNC entropies are more negative than MD by approximately 0.064–$0.093 k_B$ per particle, so HNC predicts entropy increments considerably better than absolute values.

Relative to their baselines, H5 and H7 raise the MD excess entropy by $0.30214 k_B$ and $0.30149 k_B$. The moderate split M5 raises it by $0.03039 k_B$ relative to U4. These increments are much larger than the quadrature and averaging-window sensitivities (Sec. IV H).

B. Pair Entropy and the Beyond-Pair Contribution

Equation (13) was evaluated directly from the saved all-particle MD trajectories. Labels with identical charge and interaction parameters are merged before forming $g_{ij}(r)$, and the finite-count ideal normalization excludes self pairs. Table V compares the resolved MD s_2 with the full excess entropy. Across the eight mixtures, s_2 accounts for approximately 80–82% of s_{ex}. The remaining $s_{\mathrm{ex}} - s_2 \simeq -0.32$ to $-0.36 k_B$ per particle is not negligible and varies with composition.

The HNC pair entropies are systematically less negative than the MD estimates by about 0.027–$0.045 k_B$ per particle. Probe–solvent RDFs from HNC and MD have similar first-shell structures, with small differences in peak height and first minimum (Fig. 1). HNC and MD pair entropies therefore agree qualitatively but not exactly.

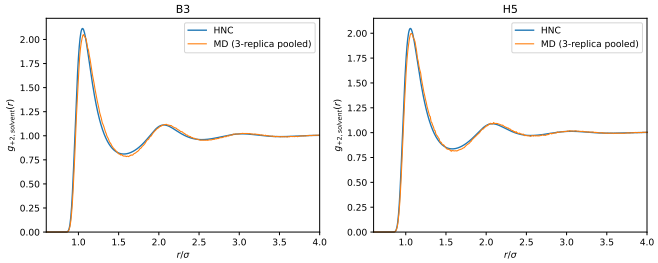


FIG. 1. HNC and pooled three-replica MD radial distribution functions for a +2 probe and the primary neutral solvent in B3 and H5. Their first coordination shells agree closely, with modest differences in the first peak and minimum.

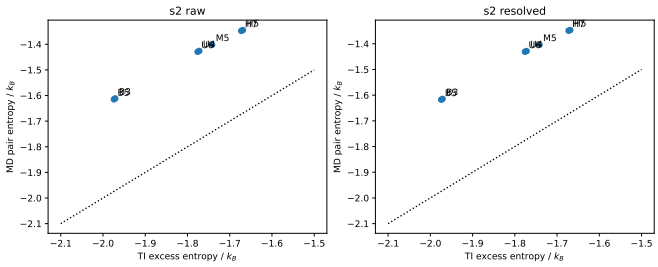


FIG. 2. MD pair entropy compared with the full excess entropy from thermodynamic integration. The left panel uses the uncorrected $\Delta r = 0.02$ RDF estimator; the right panel uses the resolved fine-core estimate. The dotted line denotes equality.

Table VI shows that pair entropy preserves most of the controlled increments: s_2 captures about 82% of the U4-to-M5 increase and 79% of the U4-to-H5 and U6-to-H7 increases. For the exact U4/L5 relabeling, both MD pair entropy and all diffusion values are identical. The pair approximation thus retains the entropy ordering, but understates the thermodynamic entropy change and leaves a composition-dependent residual (Fig. 2).

C. Label Count, Ideal Mixing, and Physical Diversification

U4 and L5 provide an exact mechanical null. Splitting the secondary solvent into two identical labels increases the formal ideal mixing entropy by $0.30 \ln 2 = 0.20794 k_B$ per particle if the labels are treated as distinguishable. Nevertheless, the diffusion coefficients are identical in every replica, and the excess-entropy difference is only $+0.00042 k_B$, interval $[-0.00219, 0.00304] k_B$. Component labels alone are not a mechanical driving variable for diffusion.

M5 and H5 have the same mole fractions and ideal mixing entropy as L5, but different interactions, so their entropy differences are interaction-induced excess changes. Adding monovalent salt to the baseline, B3→B5, increases the ideal mixing contribution by $0.38787 k_B$ but slightly *lowers* the excess entropy, $\Delta s_{\mathrm{ex}} = -0.00320 k_B$, interval $[-0.00543, -0.00097] k_B$. Neither mean diffusion

TABLE II. Dense-state HNC excess entropy and replicated MD means at $T^* = 2$, $\rho^* = 0.7$, $N = 4000$. Diffusion coefficients are in σ^2/τ; $D_{\mathrm{probe}} = (D_{+2} + D_{-2})/2$. MD uses the primary 20–100τ MSD fitting window. L5 is an exact mechanical duplicate of U4 and is excluded from entropy-scaling fits. Confidence intervals for the primary comparisons are reported in Table IV; absolute-state intervals are in Table III.

State	HNC s_{ex}/k_B	MD s_{ex}/k_B	D_{net}	D_{probe}	D_{+2}	D_{-2}
B3	-2.06350	-1.97101	0.22253	0.20677	0.21196	0.20158
B5	-2.06770	-1.97421	0.22279	0.21047	0.20879	0.21214
U4	-1.84340	-1.77209	0.25945	0.23684	0.23913	0.23455
L5	-1.84340	-1.77166	0.25945	0.23684	0.23913	0.23455
M5	-1.81020	-1.74170	0.26720	0.24038	0.23937	0.24139
H5	-1.73270	-1.66887	0.28401	0.24825	0.24764	0.24885
U6	-1.84780	-1.77603	0.26018	0.23572	0.23742	0.23401
H7	-1.73700	-1.67272	0.28196	0.24795	0.24634	0.24956

TABLE III. Absolute-state nominal 95% Student-t intervals across three independent replicas. Because the states are paired by replica, interval overlap here is not a test of the paired differences in Table IV. These intervals exclude systematic uncertainty.

State	s_{ex}/k_B	D_{net}	D_{probe}
B3	$[-1.97339, -1.96864]$	$[0.21959, 0.22546]$	$[0.20364, 0.20991]$
B5	$[-1.97657, -1.97186]$	$[0.22095, 0.22463]$	$[0.20626, 0.21468]$
U4	$[-1.77342, -1.77076]$	$[0.25884, 0.26005]$	$[0.23336, 0.24032]$
L5	$[-1.77421, -1.76912]$	$[0.25884, 0.26005]$	$[0.23336, 0.24032]$
M5	$[-1.74422, -1.73917]$	$[0.26358, 0.27082]$	$[0.22969, 0.25107]$
H5	$[-1.66936, -1.66838]$	$[0.28214, 0.28588]$	$[0.23921, 0.25728]$
U6	$[-1.77667, -1.77539]$	$[0.25480, 0.26557]$	$[0.21782, 0.25361]$
H7	$[-1.67584, -1.66960]$	$[0.27882, 0.28511]$	$[0.23406, 0.26183]$

(+0.12%) nor probe diffusion (+1.79%) changes significantly. More charged components therefore do not by themselves raise the excess entropy or accelerate the probes.

D. Divalent-Ion Enhancement

The largest design changes clearly increase divalent-ion mobility. H5/B3 gives +20.05%, interval [14.30, 26.09]%, and H7/B5 gives +17.79%, interval [11.17, 24.81]%. Both signs increase separately: +16.8% for +2 and +23.4% for −2 in H5/B3, and +18.0% and +17.6% in H7/B5, with all four paired intervals excluding zero. These comparisons also alter the mean interaction strength and local friction, so the entropy is a correlate rather than an isolated causal variable.

The controlled diversification gives a smaller response. U4→H5 preserves charged composition, ionic strength, masses, nominal diameters, and mean bare pair amplitude. Mean diffusion increases by 9.47%, interval [8.96, 9.97]%, whereas probe diffusion increases by 4.81%, interval [0.60, 9.19]%. With monovalent background, U6→H7 increases mean diffusion by 8.37%, interval [7.31, 9.44]%, and probe diffusion by 5.20%, interval [−6.14, 17.92]%. The moderate split raises mean diffusion by 2.99% with an unresolved 1.48% probe increase. The ions respond less strongly than the solvent-dominated mean, but with three replicas the size of their response remains uncertain. The extended sampling below resolves it.

E. Extended Sampling: The Controlled Ion Response

The six long replicas resolve the controlled probe response (Table VII). In the primary window, U4→H5 increases probe diffusion by 5.06%, interval [3.99, 6.13]%, and U6→H7 by 5.90%, interval [4.67, 7.14]%. Both signs accelerate separately in both comparisons. The point estimates agree with the original three-replica values, while the intervals are several times narrower. The probes respond about 0.6 times as strongly as the mean fluid, which accelerates by 8.95% and 9.17%, respectively.

The H7/U6 response is stable over all five fitting windows, from 5.5 to 5.9%, and remains resolved up to 200–1000τ (Table VIII and Fig. 3). The H5/U4 response decreases from 5.4% in the shortest window to about 2.7% beyond 100τ. This drift comes from U4, whose probe diffusivity rises by about 2% between the shortest and longest windows, while that of H5 and the mean diffusivities of all states are window independent. The probes in the uniform weak solvent therefore reach the diffusive regime more slowly. We report the controlled H5/U4 response as +3 to +5%, positive in every window. The longest windows contain the fewest independent time intervals and have correspondingly wide intervals.

The shear viscosity explains the different responses of the mean fluid and the probes (Table IX). Diversification lowers η by 7.6% from U4 to H5 and by 7.1% from U6 to H7. The mean self-diffusion follows almost exactly: $D_{\mathrm{net}}\eta$ changes by only +0.7% and +1.4%, consistent

TABLE IV. Paired composition changes, numerator relative to denominator. Brackets are nominal 95% Student-t intervals from three independent replica pairs (two degrees of freedom), without multiple-comparison correction. Diffusion changes are $100[\exp(\overline{\ln(D_a/D_b)}) - 1]$ and intervals are transformed in the same way. Entropy differences are per particle in k_B. Intervals quantify replica sampling, excluding estimator and finite-size bias.

Comparison	$\Delta s_{\mathrm{ex}}/k_B$ [95% CI]	ΔD_{net} (%) [95% CI]	$\Delta D_{\mathrm{probe}}$ (%) [95% CI]
L5/U4	+0.00042 [−0.00219, +0.00304]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]
M5/U4	+0.03039 [+0.02727, +0.03352]	+2.99 [+1.38, +4.62]	+1.48 [−1.57, +4.64]
H5/U4	+0.10322 [+0.10209, +0.10434]	+9.47 [+8.96, +9.97]	+4.81 [+0.60, +9.19]
H5/B3	+0.30214 [+0.29927, +0.30501]	+27.63 [+25.40, +29.90]	+20.05 [+14.30, +26.09]
H7/U6	+0.10331 [+0.09990, +0.10672]	+8.37 [+7.31, +9.44]	+5.20 [−6.14, +17.92]
H7/B5	+0.30149 [+0.29823, +0.30476]	+26.56 [+25.99, +27.13]	+17.79 [+11.17, +24.81]
B5/B3	−0.00320 [−0.00543, −0.00097]	+0.12 [−1.79, +2.06]	+1.79 [−1.37, +5.04]

TABLE V. Full MD thermodynamic-integration excess entropy and directly calculated all-species pair entropy at $T^* = 2$, $\rho^* = 0.7$. Pair entropy uses the fine-core resolved RDF and is compared with TI state by state. The fraction is the ratio of state means and is descriptive. The residual is $s_{\mathrm{ex}}^{\mathrm{TI}} - s_2^{\mathrm{MD}}$; it contains higher-order correlations and estimator differences.

State	TI s_{ex}/k_B	MD s_2/k_B	s_2/s_{ex}	Residual $(s_{\mathrm{ex}} - s_2)/k_B$
B3	-1.9710	-1.6146	81.9%	-0.3565
B5	-1.9742	-1.6177	81.9%	-0.3565
U4	-1.7721	-1.4283	80.6%	-0.3438
L5	-1.7717	-1.4283	80.6%	-0.3434
M5	-1.7417	-1.4035	80.6%	-0.3382
H5	-1.6689	-1.3466	80.7%	-0.3223
U6	-1.7760	-1.4310	80.6%	-0.3450
H7	-1.6727	-1.3490	80.6%	-0.3237

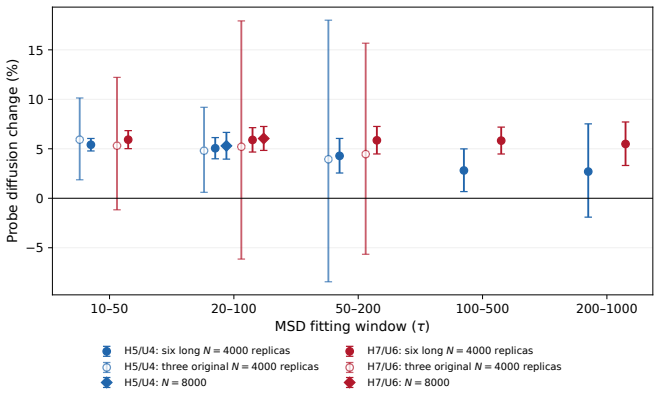


FIG. 3. Controlled probe-diffusion changes and nominal paired 95% intervals as a function of the MSD fitting window. Filled circles: six long $N = 4000$ replicas; open circles: the original three replicas; diamonds: $N = 8000$ (primary window).

with Stokes–Einstein behavior of the bulk. The probe product $D_{\mathrm{probe}}\eta$ instead decreases by 2.9% and 1.7%. Interaction diversity makes the solvent more fluid, and the divalent ions are only partly carried along by this change; their local solvation partly decouples them from the bulk viscosity.

Box size does not affect these conclusions (Table X). At $N = 8000$, all diffusivities are 0.8–1.3% larger than at $N = 4000$, close to the 1.2–1.4% predicted by Eq. (20) from the measured viscosities. After that correction, the two sizes agree within about 0.4%. The controlled probe responses are unchanged: +5.30%, interval [3.96, 6.66]%, for H5/U4 and +6.04%, interval [4.84, 7.25]%, for H7/U6 at $N = 8000$. The thermodynamics is equally insensitive to system size: the compressibility factor and the potential energy per particle of all five states agree between the two sizes within 0.03%. The mean $N = 8000$ shear viscosities differ from the $N = 4000$ values by at most 3.3%, well inside their own 95% intervals (half-widths of 5–14% from four replicas, since the Green–Kubo noise does not decrease with N). Apart from the expected hydrodynamic shift of the diffusivities, doubling the system size therefore produces no relevant change in any structural, thermodynamic, or transport result reported here.

F. Higher Divalent Concentration

Raising the divalent fraction to 0.075 per sign leaves the entropy and diffusion contrasts essentially unchanged (Tables XI and XII, Fig. 4). The controlled diversification raises the excess entropy by $0.1027k_B$ (H5/U4) and $0.1041k_B$ (H7/U6), compared with $0.1032k_B$ and $0.1033k_B$ at $x_{\pm 2} = 0.03$. Relative to the baselines, H5 and H7 gain $0.3014k_B$ and $0.3015k_B$. HNC gives $0.1108k_B$ and $0.3306k_B$ for the H5 increments, again 8–10% too large, and its absolute entropies are again more negative than MD, by 0.064–$0.093k_B$. The label-only

TABLE VI. Paired differences in full TI excess entropy, MD pair entropy, beyond-pair residual, and the individual probe diffusivities. Entropies are per particle in k_B. Diffusion entries are point percent changes from paired log ratios; their nominal 95% intervals are discussed in the text. The identical-label split is an exact mechanical null.

Comparison	$\Delta s_{\mathrm{ex}}^{\mathrm{TI}}/k_B$ [95% CI]	$\Delta s_2^{\mathrm{MD}}/k_B$ [95% CI]	$\Delta(s_{\mathrm{ex}}-s_2)/k_B$ [95% CI]	ΔD_{+2}	ΔD_{-2}
L5/U4	+0.00042 [−0.00219, +0.00304]	+0.00000 [+0.00000, +0.00000]	+0.00042 [−0.00219, +0.00304]	+0.0%	+0.0%
M5/U4	+0.03039 [+0.02727, +0.03352]	+0.02485 [+0.02300, +0.02669]	+0.00555 [+0.00247, +0.00862]	+0.0%	+2.9%
H5/U4	+0.10322 [+0.10209, +0.10434]	+0.08170 [+0.08077, +0.08262]	+0.02152 [+0.02057, +0.02247]	+3.6%	+6.1%
H7/U6	+0.10331 [+0.09990, +0.10672]	+0.08202 [+0.08154, +0.08250]	+0.02129 [+0.01756, +0.02502]	+3.7%	+6.6%
H5/B3	+0.30214 [+0.29927, +0.30501]	+0.26795 [+0.26583, +0.27007]	+0.03419 [+0.02947, +0.03891]	+16.8%	+23.4%
H7/B5	+0.30149 [+0.29823, +0.30476]	+0.26870 [+0.26701, +0.27038]	+0.03280 [+0.03055, +0.03505]	+18.0%	+17.6%
B5/B3	−0.00320 [−0.00543, −0.00097]	−0.00312 [−0.00545, −0.00080]	−0.00008 [−0.00387, +0.00372]	-1.5%	+5.2%

TABLE VII. Long $N = 4000$ paired diffusion changes at the primary 20–100τ MSD fitting window. Six independent matched replicas of 5000τ each; brackets are nominal 95% paired Student-t intervals (five degrees of freedom), without multiplicity correction. Changes are $100[\exp(\overline{\ln(D_a/D_b)}) - 1]$ in percent.

Comparison	D_{net} [CI] (%)	D_{probe} [CI] (%)	D_{+2} [CI] (%)	D_{-2} [CI] (%)
H5/U4	+8.95 [+8.62, +9.28]	+5.06 [+3.99, +6.13]	+6.02 [+3.99, +8.08]	+4.10 [+2.84, +5.38]
H7/U6	+9.17 [+8.98, +9.37]	+5.90 [+4.67, +7.14]	+5.36 [+4.46, +6.27]	+6.44 [+4.56, +8.35]
H5/B3	+27.65 [+27.31, +28.00]	+18.66 [+17.21, +20.13]	+19.85 [+17.92, +21.81]	+17.47 [+15.93, +19.03]
U4/B3	+17.17 [+16.61, +17.73]	+12.95 [+11.94, +13.96]	+13.05 [+11.09, +15.05]	+12.84 [+11.75, +13.94]

control remains exact: L5 and U4 have identical transport in every replica and $\Delta s_{\mathrm{ex}} = +0.0002 k_B$, interval $[-0.0012, 0.0015] k_B$. Monovalent salt addition again lowers the excess entropy slightly, by $0.0034 k_B$, and leaves the probe diffusion unchanged (+0.12%).

The controlled probe response is the same as at the lower concentration. U4→H5 increases probe diffusion by 5.18%, interval [3.65, 6.73]%, and U6→H7 by 6.01%, interval [4.63, 7.41]%, against 5.06% and 5.90% at $x_{\pm 2} = 0.03$. Both signs again accelerate separately, by 4.9 and 5.5% for H5/U4 and by 5.8 and 6.2% for H7/U6. The mean fluid accelerates by 8.7 and 9.1%, so the probes again respond about 0.6 times as strongly. The moderate split M5 now resolves a small probe gain of 1.70%, interval [0.94, 2.47]%. The design comparisons H5/B3 and H7/B5 give +19.1% and +19.5%, with intervals [17.2, 21.1]% and [17.3, 21.6]%. Diversification lowers the viscosity by 9.2%, interval [4.9, 13.3]%, from U4 to H5. Within this uncertainty, the mean diffusion again follows the viscosity ($D_{\mathrm{net}}\eta$ changes by −1.3%), while the probes lag ($D_{\mathrm{probe}}\eta$ changes by −4.5%). The H7/U6 viscosity change of −5.7% is not resolved with four replicas.

The additional ions shift all states nearly uniformly. The excess entropy decreases by $0.016 k_B$ per particle in every state, which HNC also predicts. The compressibility factor falls by 0.020 and the potential energy per particle by 0.078–0.082. The probe diffusivities drop by only 0.5–1.3%, and the viscosities are unchanged within their uncertainties. The ions differ from the primary solvent only in charge, so at $T^* = 2$ the added Coulomb coupling barely affects the dynamics.

Table XIII reports the ionic conductivity. The running Green–Kubo integral reaches its plateau within 2–5τ, and its values at 2, 5, 10, and 20τ agree within their uncertainties. In every state, $\sigma/\sigma_{\mathrm{NE}}$ lies between 0.69 and 0.74. Pooled over the 28 independent replicas, it is 0.727, interval [0.708, 0.746]. Correlated motion of oppositely charged ions therefore lowers the conductivity by about 27% relative to independent carriers, by the same amount in every composition: no paired change in $\sigma/\sigma_{\mathrm{NE}}$ is resolved. The conductivity thus follows the number of carriers and their self-diffusion. Adding monovalent salt raises it by 18.8%, interval [6.6, 32.4]% (B5/B3), compared with 17.7% from Nernst–Einstein. The controlled diversification predicts +5.2% and +6.1% from Nernst–Einstein. The measured changes, −1.3%, interval [−10.4, 8.6]%, and +4.0%, interval [−9.7, 19.8]%, are consistent with these predictions, but Green–Kubo conductivity from four replicas cannot resolve a change of this size. For electrolyte design, the conductivity gain available from interaction diversity at fixed mean pair amplitude is therefore bounded by the 5–6% gain in ion self-diffusion. Changing the solvent interaction strength itself gives 13–19%.

G. Rosenfeld-Like Correlation and Its Limits

Figure 5 shows a positive entropy–mobility trend for both the mean and the divalent probes. All masses, temperatures, and densities are equal, so $D_i^R = D_i \rho^{1/3}/\sqrt{T}$ differs from D_i by a common constant. Unweighted fits of composition means to

$$\ln D_i^R = \ln A_i + B_i s_{\mathrm{ex}}/k_B \tag{22}$$

give $B_{\mathrm{net}} = 0.80$ and $B_{\mathrm{probe}} = 0.62$ for B3/U4/M5/H5, and 0.78 and 0.55 for B5/U6/H7, all with $R^2 > 0.99$. These fits contain only four or three composition means and do not incorporate uncertainty in both coordinates;

TABLE VIII. Long $N = 4000$ probe-diffusion changes by MSD fitting window, with nominal 95% paired Student-t intervals from six matched replicas. The 20–100τ window is primary. U4 and H5 columns are means of six absolute replica diffusivities in LJ units and show the source of the H5/U4 window dependence.

Window	H5/U4 (%) [CI]	H7/U6 (%) [CI]	U4 D_{probe}	H5 D_{probe}
10–50τ	+5.41 [+4.78, +6.04]	+5.92 [+5.01, +6.84]	0.23642	0.24921
20–100τ	+5.06 [+3.99, +6.13]	+5.90 [+4.67, +7.14]	0.23709	0.24908
50–200τ	+4.28 [+2.55, +6.05]	+5.86 [+4.48, +7.26]	0.23821	0.24842
100–500τ	+2.81 [+0.68, +4.99]	+5.82 [+4.47, +7.19]	0.24065	0.24742
200–1000τ	+2.70 [−1.91, +7.52]	+5.49 [+3.31, +7.71]	0.24129	0.24780

TABLE IX. Long $N = 4000$ absolute transport properties at $T = 2$, $\rho = 0.7$ from six independent 5000τ replicas. Brackets are nominal 95% Student-t intervals; η is the stress-correlation plateau. $D\eta$ is formed within each replica before averaging. The last column applies the Yeh–Hummer correction to each probe diffusivity. All values use LJ units.

State	D_{net} [CI]	D_{probe} [CI]	$D_{+2} - D_{-2}$ [CI]	η [CI]	$D_{\mathrm{net}}\eta$	$D_{\mathrm{probe}}\eta$	$D^{\mathrm{YH}}_{\mathrm{probe}}$ [CI]
B3	0.22223 [0.22155, 0.22291]	0.20991 [0.20859, 0.21123]	−0.00106 [−0.00359, 0.00146]	1.24679 [1.21710, 1.27647]	0.27709	0.26170	0.22343 [0.22190, 0.22495]
U4	0.26039 [0.25975, 0.26103]	0.23709 [0.23567, 0.23851]	−0.00075 [−0.00508, 0.00357]	1.08932 [1.06026, 1.11837]	0.28365	0.25826	0.25256 [0.25098, 0.25414]
H5	0.28368 [0.28317, 0.28419]	0.24908 [0.24735, 0.25081]	0.00375 [0.00129, 0.00620]	1.00688 [0.96547, 1.04829]	0.28565	0.25078	0.26582 [0.26380, 0.26784]
U6	0.25929 [0.25890, 0.25968]	0.23537 [0.23373, 0.23700]	0.00107 [−0.00310, 0.00524]	1.08008 [1.04153, 1.11864]	0.28005	0.25426	0.25097 [0.24981, 0.25213]
H7	0.28307 [0.28248, 0.28366]	0.24925 [0.24756, 0.25093]	−0.00141 [−0.00414, 0.00132]	1.00292 [0.96407, 1.04176]	0.28389	0.24998	0.26605 [0.26434, 0.26777]

the high R^2 is descriptive rather than an independent validation of a universal law. Replacing s_{ex} by the MD pair entropy preserves the positive trends, with R^2 between 0.989 and 1.000 for the mean, pooled-probe, and separate-sign fits (Fig. 6).

The controlled local response differs from the family fits. With the long replicas, $\Delta \ln D/\Delta(s_{\mathrm{ex}}/k_B)$ for H5/U4 is approximately 0.83 for the mean and 0.48 for the probes; H7/U6 gives approximately 0.85 and 0.56. The probe response is thus weaker than the mean response under the same intervention. Mean self-diffusion assigns only 6% weight to the divalent particles and cannot determine their mobility. The results support a restricted entropy–mobility association for both observables, while neither component count nor a single bulk entropy determines species mobility.

H. Estimator Sensitivity and Homogeneity

Charge-sign symmetry is a convergence diagnostic. All five states studied in the extended campaign are invariant under global charge reversal, so the long-time self-diffusion of the two signs must coincide. In the original three replicas, B3 gave $D_{+2} - D_{-2} = 0.0104$, interval $[0.0029, 0.0179]$. The six long replicas give -0.0011, interval $[-0.0036, 0.0015]$, and the pooled nine replicas give $+0.0028$, interval $[-0.0020, 0.0075]$. The earlier asymmetry was therefore a sampling effect. In the other states the sign difference is consistent with zero in every window. The only exception is H5 in the three shortest windows, with 0.0037 (1.5%) in the primary window. These windows reuse the same trajectories, so they are not independent tests. The difference is small and vanishes in the two longest windows, so we regard it as a residual sampling fluctuation without multiplicity correction.

Switching from cubic to shape-preserving interpolation changes the dense-state entropy by at most $0.00032k_B$, and retaining all or 75% of production instead of the final half changes it by at most $0.00165k_B$. These sensitivities are small relative to the $0.103k_B$ diversification increments, although they do not bound the error below the lowest simulated density. Low-wavevector species concentration fluctuations remain of order unity, approximately 0.48–1.80 for individual replica/species values, with no enhancement indicating incipient demixing.

V. CONCLUSIONS

The replicated calculations establish a positive excess-entropy–mobility association in the selected solvent-design families, with clear limits on its interpretation.

1. **Component count alone has no transport effect.** An exact label-only split raises the formal ideal mixing entropy while leaving diffusion identical. Monovalent salt addition slightly lowers the excess entropy and gives no resolved divalent speedup.

2. **HNC-guided interaction changes produce higher-excess-entropy, faster-ion liquids.** MD confirms entropy increases of approximately $0.30k_B$ and divalent diffusion gains of 18–20% relative to the baselines, with both signs accelerating. HNC predicts the entropy increments to within about 7%. These are design comparisons, not isolated entropy interventions.

3. **Controlled diversification gives a weaker ion response.** At fixed charged composition and mean bare pair amplitude, the excess entropy rises by approximately $0.103k_B$, mean diffusion by 8–9%, and

TABLE X. Finite-size transport comparison at $T = 2$, $\rho = 0.7$, with six $N = 4000$ and up to four $N = 8000$ replicas per state. Absolute diffusivities are means in LJ units. Raw and Yeh–Hummer-corrected size ratios are percent changes with unpaired Welch 95% intervals. The predicted size ratio uses the $N = 4000$ viscosity and $\Delta D = 2(2.837297)/(6\pi\eta)(1/L_{4000} - 1/L_{8000})$, $L_N = (N/0.7)^{1/3}$. The final two rows instead give paired probe changes within each size, with Student-t 95% intervals. Missing results are shown as em dashes.

State	Observable	$N = 4000$	$N = 8000$	Raw ratio (%) [CI]	YH predicted (%)	YH ratio (%) [CI]
B3	D_{net}	0.22223	0.22474	+1.13 [+0.81, +1.45]	+1.25	−0.07 [−0.64, +0.49]
B3	D_{probe}	0.20991	0.21268	+1.32 [+0.67, +1.97]	+1.33	+0.04 [−0.95, +1.03]
U4	D_{net}	0.26039	0.26289	+0.96 [+0.59, +1.33]	+1.22	−0.27 [−0.80, +0.26]
U4	D_{probe}	0.23709	0.23927	+0.92 [+0.17, +1.67]	+1.35	−0.42 [−1.18, +0.34]
H5	D_{net}	0.28368	0.28598	+0.81 [+0.62, +1.00]	+1.22	−0.39 [−0.65, −0.13]
H5	D_{probe}	0.24908	0.25194	+1.15 [+0.14, +2.17]	+1.39	−0.22 [−1.12, +0.68]
U6	D_{net}	0.25929	0.26234	+1.17 [+0.87, +1.48]	+1.24	+0.09 [−0.32, +0.50]
U6	D_{probe}	0.23537	0.23798	+1.11 [+0.23, +2.00]	+1.37	−0.07 [−0.73, +0.60]
H7	D_{net}	0.28307	0.28620	+1.11 [+0.75, +1.47]	+1.22	−0.22 [−0.72, +0.27]
H7	D_{probe}	0.24925	0.25236	+1.25 [+0.38, +2.13]	+1.39	−0.26 [−1.20, +0.69]
H5/U4	$\Delta D_{\mathrm{probe}}$ (%)	+5.06 [+3.99, +6.13]	+5.30 [+3.96, +6.66]	—	—	—
H7/U6	$\Delta D_{\mathrm{probe}}$ (%)	+5.90 [+4.67, +7.14]	+6.04 [+4.84, +7.25]	—	—	—

TABLE XI. Paired contrasts at $x_{+2} = x_{-2} = 0.075$, $N = 4000$, four matched replicas of 3000τ each, $\rho^* = 0.7$, $T^* = 2$. Brackets give 95% confidence intervals. Percentages are $100[\exp(\ln(A/B)) - 1]$; σ is the Green–Kubo conductivity from the 5–10τ plateau. HNC entropy increments are evaluated at the same divalent fraction.

Contrast	MD $\Delta s_{\mathrm{ex}}/k_B$ [CI]	HNC $\Delta s_{\mathrm{ex}}/k_B$	D_{net} (%) [CI]	D_{probe} (%) [CI]	η (%) [CI]	σ (%) [CI]
H5/U4	+0.1027 [+0.1014, +0.1039]	+0.1108	+8.67 [+7.56, +9.79]	+5.18 [+3.65, +6.73]	−9.19 [−13.27, −4.91]	−1.33 [−10.37, +8.63]
M5/U4	+0.0301 [+0.0285, +0.0317]	+0.0332	+2.49 [+1.54, +3.45]	+1.70 [+0.94, +2.47]	−2.31 [−10.78, +6.96]	−1.84 [−12.67, +10.33]
L5/U4	+0.0002 [−0.0012, +0.0015]	+0.0000	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]
H7/U6	+0.1041 [+0.1031, +0.1051]	+0.1109	+9.08 [+8.76, +9.40]	+6.01 [+4.63, +7.41]	−5.69 [−17.44, +7.74]	+3.97 [−9.74, +19.77]
H5/B3	+0.3014 [+0.2996, +0.3032]	+0.3306	+27.34 [+26.38, +28.31]	+19.09 [+17.15, +21.06]	−20.71 [−25.53, −15.57]	+12.92 [−4.09, +32.96]
H7/B5	+0.3015 [+0.3005, +0.3026]	+0.3306	+27.99 [+27.34, +28.64]	+19.46 [+17.32, +21.64]	−18.69 [−28.36, −7.70]	+17.35 [+5.44, +30.60]
U4/B3	+0.1987 [+0.1968, +0.2006]	—	+17.18 [+16.62, +17.75]	+13.23 [+12.76, +13.70]	−12.69 [−20.34, −4.29]	+14.44 [−7.04, +40.89]
B5/B3	−0.0034 [−0.0043, −0.0025]	−0.0044	−0.32 [−0.78, +0.14]	+0.12 [−0.60, +0.85]	+0.79 [−6.25, +8.35]	+18.80 [+6.62, +32.38]

probe diffusion by 5–6%, resolved with and without monovalent background and independent of box size. The mean diffusion tracks the accompanying drop in shear viscosity, whereas the probes are only partly carried along.

4. **Rosenfeld-like trends extend to both probe signs but do not establish a universal law.** Separate fits of D_{+2} and D_{-2} against either the excess entropy or the MD pair entropy have positive slopes within the selected families, while the controlled local slopes for the probes are about 0.6 times those of the mean.

5. **Pair entropy captures most, but not all, of the excess entropy.** The all-species s_2 accounts for approximately 80% of s_{ex} and of its controlled increments, leaving a composition-dependent beyond-pair residual of about -0.32 to $-0.36 k_B$ per particle.

6. **The conclusions hold at an electrolyte-like divalent content.** At $x_{\pm 2} = 0.075$ (total 0.15), the entropy increments, the 5–6% controlled probe response, and the 19% design gains agree with the $x_{\pm 2} = 0.03$ values within their uncertainties. The ionic conductivity is a composition-independent 0.73 of its Nernst–Einstein value, so diversification raises it no more than it raises the ion self-diffusion.

The partial decoupling of the probes from the bulk viscosity points to local solvation as the controlling factor. Probe–solvent coordination, association lifetimes, and exchange kinetics, together with several state points at fixed composition, would test whether weaker local caging rather than a bulk entropy coordinate controls the ion mobility.

ACKNOWLEDGMENTS

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Appendix A: Two-Phase Thermodynamic Estimate of the Excess Entropy

The two-phase thermodynamic (2PT) method of Lin, Blanco, and Goddard [21] estimates the entropy of a liquid from its vibrational density of states (DoS). The DoS is

TABLE XII. Paired changes at $x_{+2} = x_{-2} = 0.03$ and 0.075; brackets are 95% confidence intervals. The 0.03 results use the completed earlier campaign. A dash means the paired estimate is unavailable. Diffusion percentages use exponentiated mean log ratios.

Contrast	Δs_{ex} 0.03	Δs_{ex} 0.075	D_{net} 0.03 (%)	D_{net} 0.075 (%)	D_{probe} 0.03 (%)	D_{probe} 0.075 (%)
H5/U4	+0.1032 [+0.1021, +0.1043]	+0.1027 [+0.1014, +0.1039]	+8.95 [+8.62, +9.28]	+8.67 [+7.56, +9.79]	+5.06 [+3.99, +6.13]	+5.18 [+3.65, +6.73]
H7/U6	+0.1033 [+0.0999, +0.1067]	+0.1041 [+0.1031, +0.1051]	+9.17 [+8.98, +9.37]	+9.08 [+8.76, +9.40]	+5.90 [+4.67, +7.14]	+6.01 [+4.63, +7.41]
H5/B3	+0.3021 [+0.2993, +0.3050]	+0.3014 [+0.2996, +0.3032]	+27.65 [+27.31, +28.00]	+27.34 [+26.38, +28.31]	+18.66 [+17.21, +20.13]	+19.09 [+17.15, +21.06]
U4/B3	—	+0.1987 [+0.1968, +0.2006]	+17.17 [+16.61, +17.73]	+17.18 [+16.62, +17.75]	+12.95 [+11.94, +13.96]	+13.23 [+12.76, +13.70]

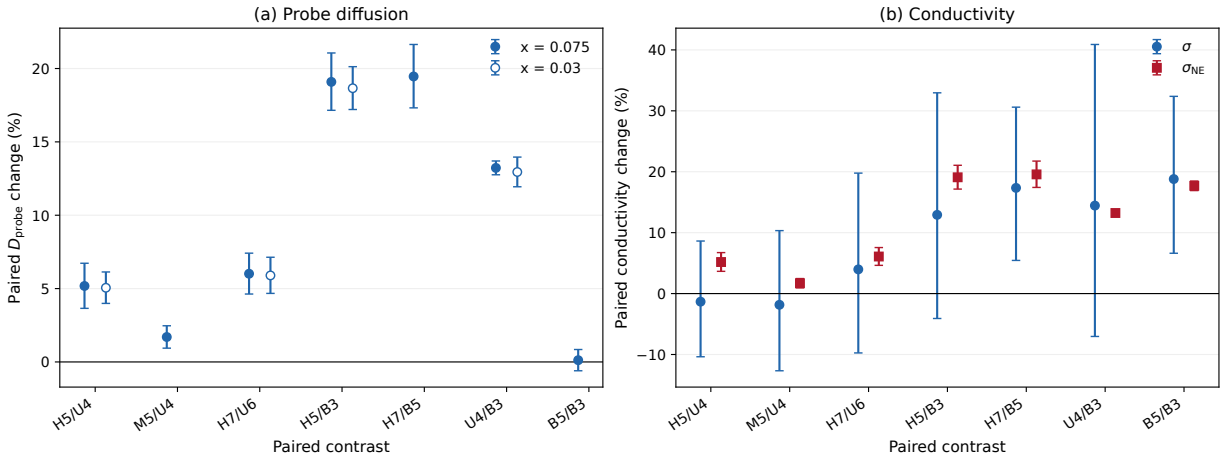


FIG. 4. Paired contrasts at $x_{+2} = x_{-2} = 0.075$ (four matched replicas, 3000τ, $N = 4000$) with nominal 95% intervals. (a) Probe self-diffusion, with the $x_{\pm 2} = 0.03$ long-run values where available. (b) Green–Kubo conductivity and its Nernst–Einstein estimate.

partitioned into a diffusive, gas-like component, modeled as a hard-sphere gas, and a solid-like remainder treated as a set of harmonic oscillators. Because it requires only a short trajectory, 2PT is an attractive alternative to thermodynamic integration. The partition is, however, a model rather than a literal population count, and in mixtures the species partial volumes must be approximated [23]. We therefore test 2PT against the integration results of Sec. IV, rather than use it as an entropy coordinate.

Density of states. For species i, let $C_i(t) = \langle \mathbf{v}(t) \cdot \mathbf{v}(0) \rangle / 3$ be the per-component VACF, so that $D_i = \int_0^\infty C_i(t)\, dt$. The DoS per particle is

$$G_i(\nu) = \frac{12\, m_i}{k_B T} \int_0^\infty C_i(t) \cos(2\pi\nu t)\, dt, \tag{A1}$$

which by equipartition satisfies $\int_0^\infty G_i(\nu)\, d\nu = 3$, and whose zero-frequency value is $G_i(0) = 12 m_i D_i/(k_B T)$.

Fluidicity. The gas-like fraction f_i of the degrees of freedom follows from the dimensionless diffusivity

$$\Delta_i = \frac{2G_i(0)}{9} \left(\frac{\pi k_B T}{m_i}\right)^{1/2} \rho_i^{1/3} \left(\frac{6}{\pi}\right)^{2/3}, \tag{A2}$$

which is the ratio of D_i to the Enskog hard-sphere diffusivity at the same density in the zero-pressure limit [21]. Here $\rho_i = 1/\bar{V}_i$ is the inverse partial volume of species i. Requiring that the gas-like particles behave as hard spheres whose packing fraction y_i and diffusivity are consistent with Δ_i gives the universal equation

$$\begin{aligned} &2\Delta_i^{-9/2} f_i^{15/2} - 6\Delta_i^{-3} f_i^{5} - \Delta_i^{-3/2} f_i^{7/2} \\ &\quad + 6\Delta_i^{-3/2} f_i^{5/2} + 2f_i - 2 = 0, \end{aligned} \tag{A3}$$

with $y_i = \Delta_i^{-3/2} f_i^{5/2}$. Equation (A3) combines two conditions that fix both f_i and the unknown hard-sphere diameter σ_{HS}. In the first, the fluidicity measures the actual diffusivity relative to the low-density (Chapman–Enskog) hard-sphere value, $D_0 = \frac{3}{8}(\rho\sigma_{\mathrm{HS}}^2)^{-1}(k_B T/\pi m_i)^{1/2}$. Sun *et al.* [22] write this condition as

$$f_i^{\delta} = D_i/D_0. \tag{A4}$$

In the second, the $f_i N_i$ gas-like particles carry all the diffusion, so their Enskog dense-gas diffusivity, $D_0(f_i\rho)/g_{\mathrm{HS}}(\sigma^+)$ with the Carnahan–Starling contact value, equals D_i/f_i. In terms of the gas packing fraction $y_i = \pi f_i \rho \sigma_{\mathrm{HS}}^3/6$, these conditions read $y_i = \Delta_i^{-3/2} f_i^{1+3\delta/2}$ and $f_i^{2/3} = (\Delta_i/2)\, y_i^{2/3}(2 - y_i)/(1 - y_i)^3$. The exponent δ is phenomenological. The original 2PT method uses $\delta = 1$, i.e. $f_i = D_i/D_0$, which recovers Eq. (A3). A larger δ raises f_i at a given Δ_i, which shifts modes from the solid to the gas component and increases the entropy. Sun *et al.* found that $\delta = 1.5$ best reproduces the equation-of-state entropy of the LJ fluid, so this value is an empirical calibration for a one-component simple liquid rather than a derived result.

TABLE XIII. Absolute-state means and 95% confidence intervals at $x_{+2} = x_{-2} = 0.075$. Conductivity σ uses the Green–Kubo 5–10τ plateau. L5 is omitted because it is mechanically identical to U4. Across the 28 replicas of the seven distinct states, pooled $\sigma/\sigma_{\mathrm{NE}} = 0.727$ [0.708,0.746].

State	s_{ex}/k_B [CI]	D_{probe} [CI]	η [CI]	σ [CI]	σ_{NE} [CI]	$\sigma/\sigma_{\mathrm{NE}}$ [CI]
B3	-1.98663 $[-1.98812, -1.98514]$	0.20728 [0.20658, 0.20799]	1.26381 [1.20948, 1.31813]	0.03198 [0.02653, 0.03742]	0.04353 [0.04338, 0.04368]	0.73447 [0.61115, 0.85779]
B5	-1.99002 $[-1.99176, -1.98827]$	0.20753 [0.20586, 0.20921]	1.27441 [1.18194, 1.36687]	0.03796 [0.03177, 0.04416]	0.05123 [0.05081, 0.05165]	0.74081 [0.62551, 0.85611]
U4	-1.78791 $[-1.78933, -1.78649]$	0.23470 [0.23346, 0.23594]	1.10372 [1.04109, 1.16634]	0.03647 [0.03344, 0.03951]	0.04929 [0.04903, 0.04955]	0.74005 [0.67636, 0.80373]
M5	-1.75779 $[-1.75872, -1.75687]$	0.23869 [0.23660, 0.24079]	1.07970 [0.95544, 1.20397]	0.03578 [0.03413, 0.03743]	0.05013 [0.04969, 0.05056]	0.71366 [0.68593, 0.74139]
H5	-1.68523 $[-1.68585, -1.68462]$	0.24686 [0.24289, 0.25083]	1.00201 [0.96538, 1.03863]	0.03598 [0.03351, 0.03844]	0.05184 [0.05101, 0.05268]	0.69414 [0.63912, 0.74917]
U6	-1.79258 $[-1.79367, -1.79148]$	0.23386 [0.23188, 0.23584]	1.09939 [0.99220, 1.20659]	0.04277 [0.03770, 0.04784]	0.05774 [0.05733, 0.05815]	0.74094 [0.64821, 0.83368]
H7	-1.68849 $[-1.68953, -1.68745]$	0.24792 [0.24425, 0.25160]	1.03632 [0.95854, 1.11411]	0.04442 [0.04106, 0.04778]	0.06125 [0.06042, 0.06209]	0.72534 [0.66577, 0.78491]

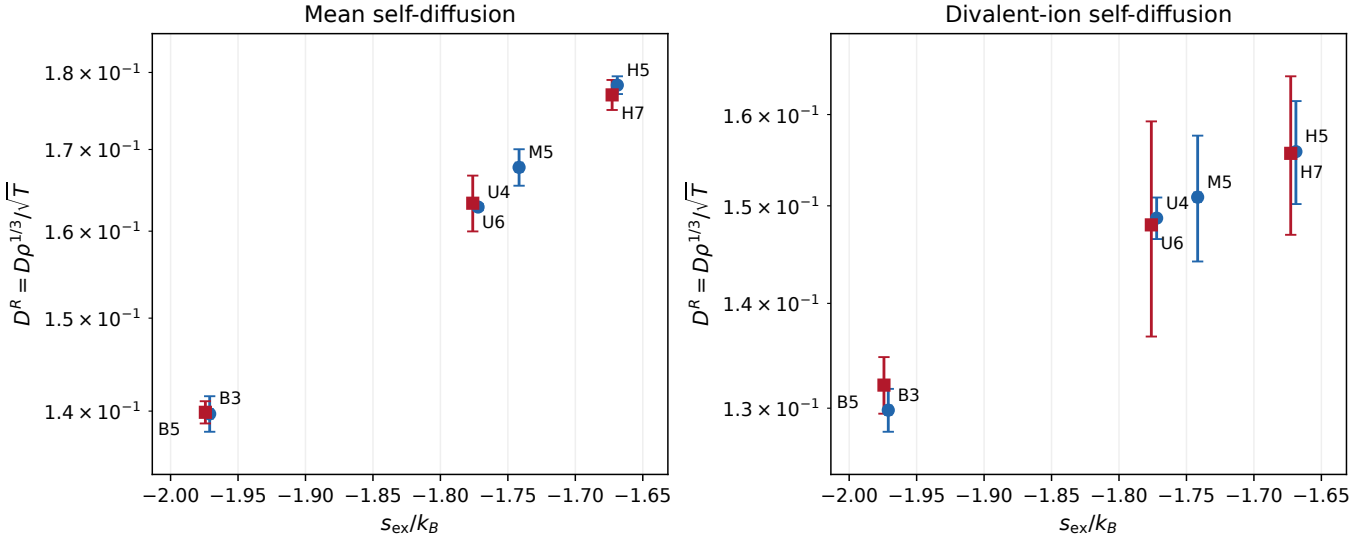


FIG. 5. Rosenfeld-reduced mean and divalent-probe self-diffusion versus MD excess entropy. Vertical axes are logarithmic. Blue circles: no monovalent background; red squares: 5% of each monovalent sign. Bars are nominal 95% absolute-state replica intervals. L5 coincides exactly with U4 in transport and is omitted.

Partition of the spectrum. The gas component is a Lorentzian with the same zero-frequency intensity,

$$G_i^g(\nu) = \frac{G_i(0)}{1 + [\pi G_i(0)\nu/(6f_i)]^2}, \qquad \int_0^\infty G_i^g \, d\nu = 3f_i, \tag{A5}$$

and the solid-like spectrum is the remainder, $G_i^s = G_i - G_i^g$, which carries $3(1 - f_i)$ modes.

Entropy. Each part is weighted by the appropriate entropy per mode,

$$\frac{s_i}{k_B} = \int_0^\infty G_i^g(\nu)\, W_i^g \, d\nu + \int_0^\infty G_i^s(\nu)\, W^s(\nu)\, d\nu. \tag{A6}$$

For classical oscillators, $W^s(\nu) = 1 - \ln(h\nu/k_BT)$. The gas weight is $W_i^g = s_i^{\mathrm{HS}}/(3k_B)$, with the Carnahan–Starling hard-sphere entropy

$$\frac{s_i^{\mathrm{HS}}}{k_B} = \frac{5}{2} + \ln\left[\left(\frac{2\pi m_i k_B T}{h^2}\right)^{3/2} \frac{1}{f_i \rho_i}\right] + \frac{y_i(3y_i - 4)}{(1 - y_i)^2}. \tag{A7}$$

The excess entropy is $s_{\mathrm{ex},i} = s_i - s^{\mathrm{id}}(\rho)$, where $s^{\mathrm{id}}/k_B = \frac{5}{2} + \ln[(2\pi m_i k_B T/h^2)^{3/2}/\rho]$ is the ideal-gas value at the total density. Because $\int G_i \, d\nu = 3$, Planck's constant cancels between Eqs. (A6) and (A7) and the ideal reference. The mixture value is $s_{\mathrm{ex}} = \sum_i x_i s_{\mathrm{ex},i}$, and the ideal mixing term cancels identically.

Implementation. We analyzed the final-100τ velocity records of all 24 replicas. Following Lai *et al.* [23], every species is assigned the equal partial volume $\bar{V}_i = 1/\rho$. We first evaluate the two published exponents, $\delta = 1$ and 1.5, without any adjustment, and then calibrate a single δ for all mixtures (see below). No other parameter, reference volume, or entropy offset is adjusted. The VACF, computed in the common COM frame, is multiplied by a Gaussian taper $e^{-(t/\tau_w)^2}$ with $\tau_w = 10\tau$. Its spectrum spans the full Nyquist band and is normalized by the measured $C_i(0)$ to exactly three modes per particle. The Lorentzian of Eq. (A5) and its contribution to Eq. (A6) are integrated analytically to infinite frequency. Following the revised formulation [22], G_i^s is kept signed where it becomes slightly negative, rather than clipped, so that the total number of modes is conserved.

The implementation passes analytic checks, including the fluidicity polynomial, unit invariance, a solvable VACF/Green–Kubo limit, the Lorentzian entropy tail, the dilute-gas limit, and cancellation of the classical reference and of label mixing. Mechanically identical labels give

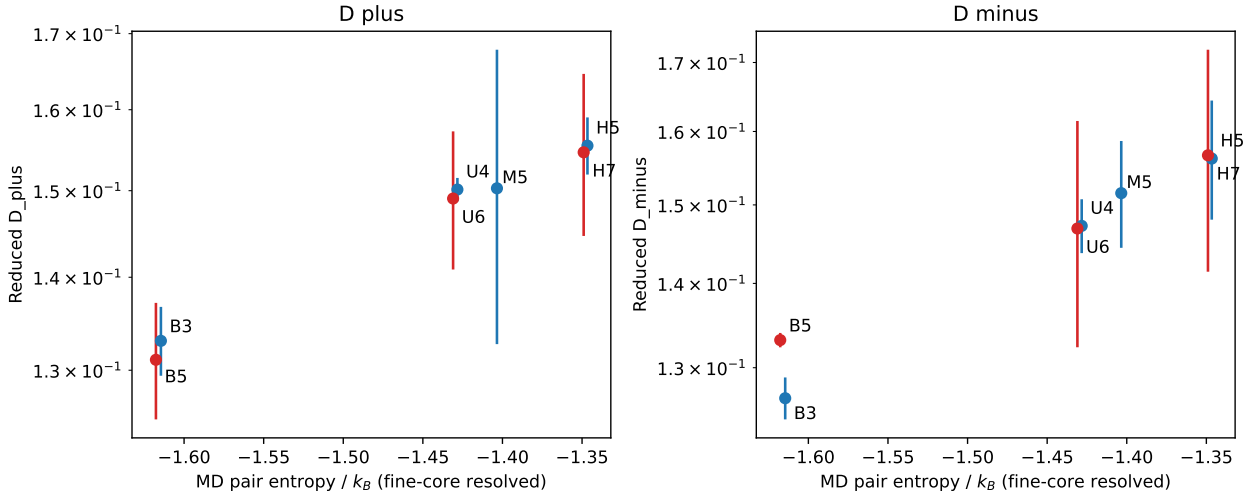


FIG. 6. Reduced self-diffusion coefficients of the +2 and −2 probes versus the all-species MD pair entropy. Error bars show nominal 95% absolute-state replica intervals and exclude finite-size and estimator systematics. Blue circles: no monovalent salt; red circles: with monovalent salt. L5 is omitted because it duplicates U4 mechanically.

identical 2PT entropies, and state-mean probe spectral diffusivities agree with the MSD coefficients within 2.6%.

Results. With the published exponents, 2PT has a systematic error. For $\delta = 1$ it underestimates s_{ex} in every mixture, by 0.019–0.105k_B per particle, while $\delta = 1.5$ overestimates it by 0.048–0.106k_B. Because δ is phenomenological, we calibrated a single value against the seven mechanically distinct mixtures by least squares on the state-mean bias, with L5 excluded as a duplicate. The fit gives $\delta^* = 1.20$, with a 95% interval of $[1.17, 1.23]$ from separate fits to the three replica sets. We adopt $\delta = 1.25$, slightly above this interval. The RMS bias is flat near the minimum, and this value reduces the RMS absolute bias from 0.067k_B ($\delta = 1$) and 0.080k_B ($\delta = 1.5$) to 0.029k_B, close to the optimum of 0.026k_B (Fig. 7). Leave-one-state-out cross-validation gives a held-out RMS bias of 0.030k_B, so the calibration is not driven by any single mixture.

A single exponent fixes the overall level but not the dependence on composition. The value of δ that reproduces TI in each mixture ranges from 1.07 (H7) to 1.32 (B5). The two baseline electrolytes need $\delta \approx 1.31$, whereas the mixtures containing the weak solvent need 1.07–1.18; there is no systematic separation between mixtures with and without monovalent salt. Figure 8 shows that the bias curves are nearly parallel in δ. Changing δ therefore shifts all mixtures together and barely changes the entropy differences between them.

The consequences for the increments are shown in Table XIV. At $\delta = 1.25$, the moderate and strong diversifications without salt are reproduced: M5/U4 gives 0.0300k_B versus 0.0304k_B, and H5/U4 gives 0.110k_B versus 0.103k_B. With monovalent background, H7/U6 is overestimated by about 30% (0.135 versus 0.103k_B), with a bias interval that excludes zero. The full design changes H5/B3 and H7/B5 are overestimated by 17–23%, because the dense baselines need a larger exponent than the diversified solvents. The label-only split L5/U4 gives zero, as it must. These increment biases depend only weakly on δ and persist under variations of taper width, record half, frequency grid, sampling interval, and partial volume.

The calibrated 2PT thus estimates absolute excess entropies to within about 0.03k_B in these mixtures. However, its residual errors depend on composition and are comparable to the controlled entropy differences of interest. Moreover, δ was fitted to the same TI data, and the fluidicity is built from the diffusion coefficient [Eq. (A4)], so 2PT cannot independently confirm an entropy–diffusion correlation. Thermodynamic integration therefore remains the entropy coordinate of the main text. 2PT is a useful, inexpensive approximate estimator once calibrated, but its species allocations are not independently measured partial molar entropies.

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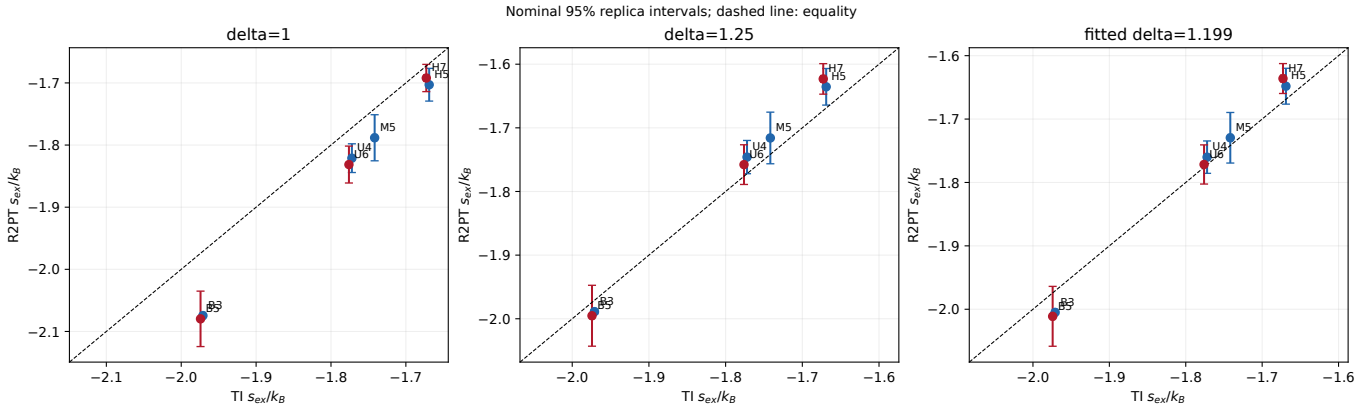


FIG. 7. Classical revised-2PT excess entropy against thermodynamic integration (TI) for the seven mechanically distinct mixtures: original partition $\delta = 1$ (left), calibrated $\delta = 1.25$ (center), and least-squares optimum $\delta^* = 1.20$ (right). L5 duplicates U4 and is omitted. Blue: no monovalent salt; red: with salt. Bars are nominal 95% replica intervals, and dashed lines denote equality.

TABLE XIV. Post-hoc global revised-2PT partition-exponent calibration. Paired excess-entropy increments are in k_B per particle; bias is 2PT minus TI, with nominal 95% Student-t intervals across three replica pairs. The seven-state unweighted fit gives $\delta^* = 1.19895$; replica-index 95% interval [1.16906,1.22816]. RMS absolute-state bias is 0.06654, 0.02931, and 0.02563 for $\delta = 1$, $\delta = 1.25$, and fitted δ^*, respectively. This calibration was not prespecified and uses the same TI states shown here.

Comparison	TI Δs_{ex}	2PT $\delta = 1.25$	Bias [95% CI]	2PT δ^*	Bias [95% CI]
M5/U4	+0.03039	+0.03004	−0.00035 [−0.05602, +0.05532]	+0.03057	+0.00018 [−0.05464, +0.05500]
H5/U4	+0.10322	+0.11043	+0.00721 [−0.01434, +0.02877]	+0.11190	+0.00868 [−0.01265, +0.03002]
H7/U6	+0.10331	+0.13467	+0.03136 [+0.01067, +0.05205]	+0.13554	+0.03223 [+0.01149, +0.05297]
H5/B3	+0.30214	+0.35329	+0.05115 [+0.01816, +0.08414]	+0.35676	+0.05462 [+0.02226, +0.08699]
H7/B5	+0.30149	+0.37213	+0.07064 [+0.04374, +0.09754]	+0.37508	+0.07358 [+0.04690, +0.10027]
B5/B3	-0.00320	-0.00645	−0.00324 [−0.04671, +0.04023]	-0.00622	−0.00301 [−0.04598, +0.03995]

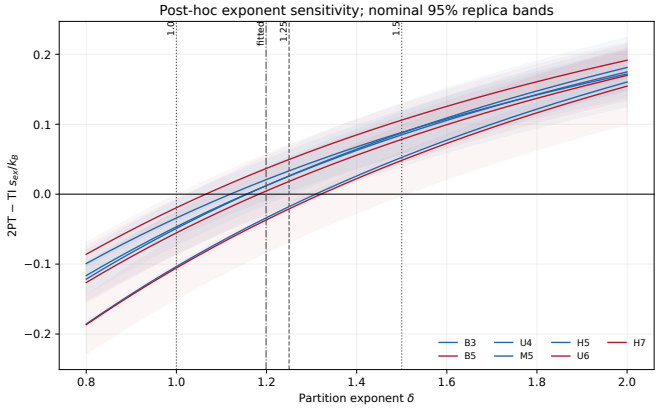


FIG. 8. Bias of the 2PT excess entropy relative to TI as a function of the partition exponent δ, with nominal 95% replica bands. Vertical lines mark $\delta = 1$, 1.5, the calibrated 1.25, and the fitted optimum. Blue: no monovalent salt; red: with salt.

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